1 **(a)** A body is released in a fluid. With the aid of a free body diagram, explain how the body falling through a fluid can reach terminal velocity.

.....

.....

.....

..... [5]

(b) A uniform rigid rod of weight 400 N is attached to a vertical beam by a hinge as shown in Fig. 1.1. The other end of the rod is fastened to a support cable. The structure is used to support a load of weight 2000 N.

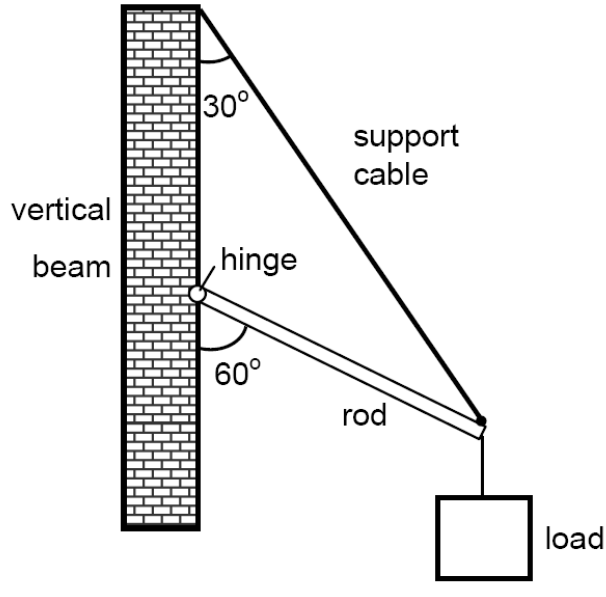

Fig. 1.1

(i) State two conditions needed for the rod to be in a state of equilibrium.

.....

..... [2]

(ii) Determine the magnitude of the tension in the support cable.

tension = ... N [3]